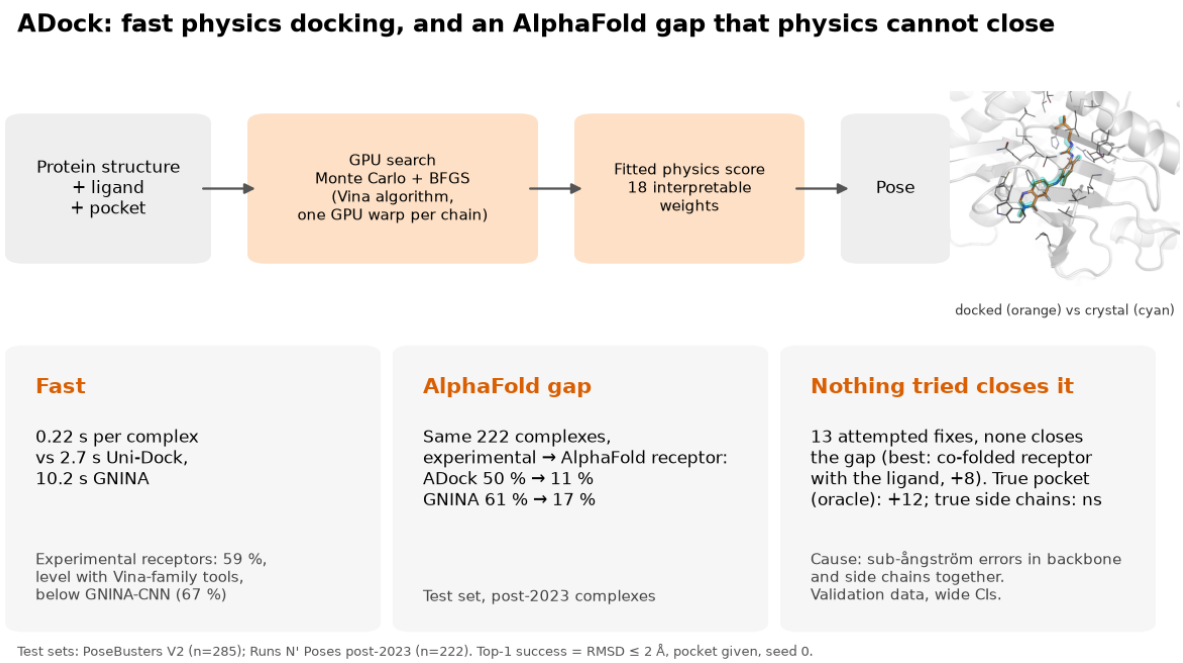
ADock — briefing

October 2026. AI-agent research project (24 h). Checked by independent agents, not yet by human experts. Details: `technical_report.pdf` *(4 pages) and* `docs/REPORT.md` *(full record).*

What was built

ADock is a fast docking program. Given a protein structure, a small molecule and the location of the binding pocket, it predicts the 3D pose in which the molecule binds.

- **How it works:** a GPU version of the classical AutoDock Vina search, scored with a refitted, interpretable physics-style function.
- **What we then asked:** why does docking fail when the protein structure comes from AlphaFold instead of an experiment?



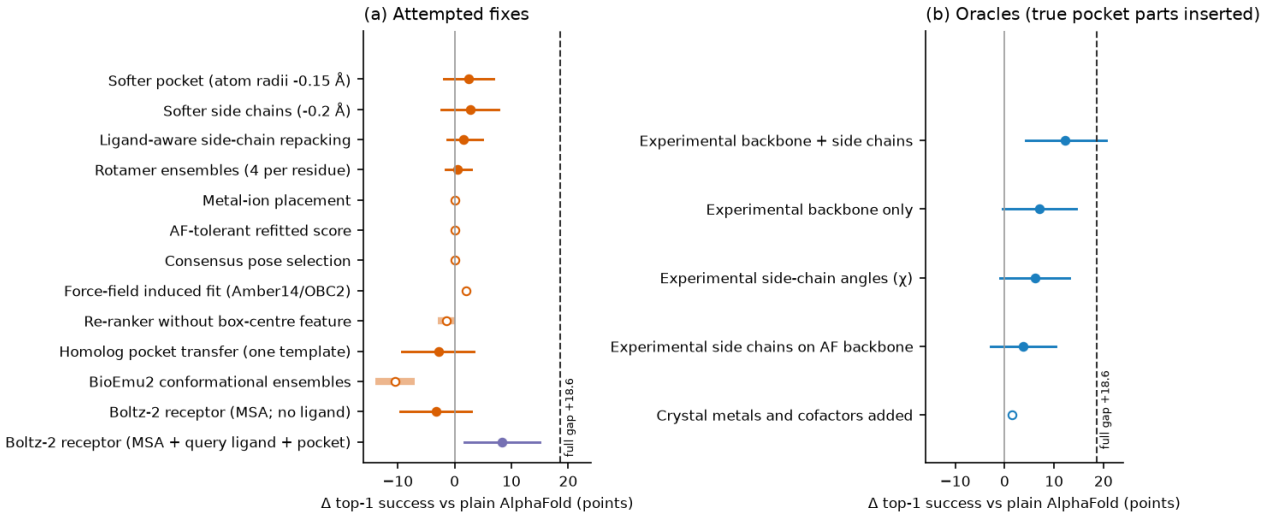
How it was tested

- **Success measure.** A prediction counts as correct if the top-ranked pose lies within 2 Å (RMSD) of the crystal pose. The pocket is given to every method in the same way.
- **Test sets,** held out until one final run:
  - **PoseBusters V2:** 285 complexes with experimental receptors.
  - **Runs N' Poses:** 222 complexes released after mid-2023, docked both into the experimental structure and into the AlphaFold model.
- **Comparison tools:** AutoDock Vina, smina, Uni-Dock (GPU Vina) and GNINA, which adds neural-network rescoring and is the strongest open tool.
- **Diagnosis data.** The causes of the AlphaFold gap were studied only on a separate validation set of 125 post-2023 complexes. One half was used for choices and the other to confirm them.
- **Pre-registration.** Three main claims and their tests were written down before the test data were opened.

Main findings

1. **Fast, with standard accuracy.**
   - ADock takes 0.2 s per complex, against 2.7 s for Uni-Dock and 10 s for GNINA.
   - On experimental structures it is as accurate as the classical tools: 59 %, against 55 % for Vina and 59 % for GNINA without its network.
   - It stays below GNINA with neural-network rescoring (67 %).
2. **AlphaFold models break physics-based docking.** On the same 222 complexes, ADock drops from 50 % on experimental to 11 % on AlphaFold receptors, and GNINA from 61 % to 17 %. This happens although the AlphaFold pockets are close to the real ones (backbone within about half an ångström).

3. **Nothing we tried closes the gap.**
   - **Thirteen fixes:** softer receptors, side-chain rebuilding, rotamer and BioEmu2 structure ensembles, metal placement, a score refitted for AlphaFold, consensus scoring, re-ranking, a full molecular-mechanics force field, pocket transfer from related structures, and co-folded receptors.
   - **Results:** each gave between −14 and +3 points, except co-folding with the actual ligand (+8). That still closes less than half the gap and uses a deep-learning model and the ligand itself.
4. **The cause is sub-ångström pocket geometry.**
   - Replacing the AlphaFold pocket by the true one recovers 12 points; true side chains or metals alone do not help significantly.
   - In AlphaFold pockets the correct pose often scores worse than wrong poses, even under a full force field. Better searching therefore cannot help.
5. **None of the three pre-registered main claims passed.**
   - ADock is not better than GNINA on AlphaFold receptors.
   - It is faster but less accurate than GNINA.
   - Its non-inferiority to the best physics tool was not significant after correction for three claims.



Validation data only. Filled dot + line: 95 % CI. Shaded bar: range of reported values (no CI). Open dot: point value. Purple: uses the query ligand and a deep-learning model.

What closes the AlphaFold gap (validation data)? Left: attempted fixes. Right: oracles that insert parts of the true pocket. The dashed line marks the full gap.

What this means

- **Experimental or high-confidence structures.** ADock suits large-scale screening where speed matters: about 66 pockets per second per GPU node, or roughly 4 node-hours per million pockets (a projection). Where per-complex accuracy matters most, GNINA is better.
- **AlphaFold models.** Physics docking into AlphaFold models of proteins without solved complexes is unreliable (about 10–20 % success for every tool). Closing the gap needs pocket models accurate well below 1 Å, or a score that tolerates such errors; neither exists yet. Reusing poses from earlier similar complexes helps only when such complexes exist.
- **Research practice.** Gains seen during method selection shrank or vanished on held-back data in nearly every experiment. Pre-registration and a held-out test changed what could be claimed.

Main limitations

- The AlphaFold diagnosis uses validation data (about 107–125 complexes) with wide confidence intervals. Some arms have only point estimates.
- The binding pocket must be known in advance.
- Timings come from a node shared with other users, so treat them as upper bounds.
- The project was run by AI agents in 24 hours. Humans have not yet reviewed the code or results.